How does disorder enter a model, a Hall current and a pump?

ResearchPhO — Solutions and marking scheme **30 points**

How to use this booklet

Each answer is followed by its marking scheme; the numbers in a marking line add up to the points of the question. (M) marks a modelling step: setting up the model, choosing the physics, interpreting a result, judging its scope. (C) marks a computation step: algebra, solving, evaluating. Most of each problem's marks go to (M). Give partial credit for a correct approximate route. Appendices A–D contain the research context; Appendix E, the toolbox, gives the smallest worked example of each tool from "Useful information". Every number in this booklet is recomputed by the answer script of the working folder.

Mark allocation

Problem	Questions	Points
1	A.1 (1.2), A.2 (1.2), B.1 (1.2), B.2 (1.0), C.1 (1.3), C.2 (1.1), C.3 (0.8), D.1 (1.0), D.2 (0.8), D.3 (0.4)	10.0
2	A.1 (0.9), A.2 (0.5), A.3 (0.8), B.1 (0.9), B.2 (1.2), B.3 (0.7), C.1 (0.9), C.2 (0.8), C.3 (1.1), D.1 (0.8), D.2 (0.9), D.3 (0.5)	10.0
3	A.1 (0.8), A.2 (0.8), B.1 (1.0), B.2 (1.0), C.1 (1.2), C.2 (0.8), C.3 (0.6), D.1 (0.6), D.2 (1.2), D.3 (1.0), D.4 (1.0)	10.0
		30.0

Problem 1: From a dirty crystal to a random Hamiltonian (10.0 points)

Part A. From an impurity to lattice parameters (2.4 points)

A.1 One impurity changes several matrix elements at once 1.2 pt

(a) The delta function gives $\delta h_{ij} = u_0\,\varphi_i(x_0)\varphi_j(x_0)$. With $\varphi_j(ja) = (\pi b^2)^{-1/4}$ and $\varphi_{j+1}(ja) = (\pi b^2)^{-1/4}e^{-a^2/(2b^2)}$:

$$x_0 = ja: \quad \delta h_{jj} = \frac{u_0}{\sqrt{\pi}\,b}, \qquad \delta h_{j,j+1} = \frac{u_0}{\sqrt{\pi}\,b}\,e^{-a^2/(2b^2)}, \qquad \delta h_{j+1,j+1} = \frac{u_0}{\sqrt{\pi}\,b}\,e^{-a^2/b^2}.$$

Midway, $\varphi_j(x_0) = \varphi_{j+1}(x_0) = (\pi b^2)^{-1/4}e^{-a^2/(8b^2)}$, so

$$x_0 = (j+\tfrac{1}{2})a: \quad \delta h_{jj} = \delta h_{j+1,j+1} = \delta h_{j,j+1} = \frac{u_0}{\sqrt{\pi}\,b}\,e^{-a^2/(4b^2)}.$$

(b) On the atom, $|\delta h_{j,j+1}/\delta h_{jj}| = e^{-a^2/(2b^2)} = e^{-8} = \boxed{3\times10^{-4}}$ (and the neighbour's on-site term is $e^{-16} \approx 10^{-7}$ of it). With orthogonalized orbitals this small ratio becomes of the order of the overlap $e^{-a^2/(4b^2)} \approx 0.02$: still small. Midway, the ratio is $\boxed{1}$. An impurity on an atom is well described by one random on-site energy: the Anderson model is adequate when impurities sit on (or near) atoms and the orbitals are compact, $b \ll a$. An impurity between atoms changes two on-site energies and the hopping between them by the same amount; a model with random on-site energies only misses half of its effect.

(c) $\delta h_{ij} = u_0\,\phi_i\phi_j$ with the vector $\phi_i = \varphi_i(x_0)$: $\delta h = u_0\,\phi\phi^T$, a matrix of rank one. All its entries are fixed by the same two random numbers (u_0, x_0). The on-site and bond terms of one impurity are therefore perfectly correlated (with the sign of u_0); drawing them as independent random numbers misrepresents the ensemble. The correct procedure samples the *source* (impurity strength and position) and computes all matrix elements from it.

Marking: (a) The matrix elements: 0.35 (C). (b) The two ratios: 0.15 (C); adequacy of the on-site model for each kind: 0.3 (M). (c) Rank one: 0.1 (C); correlated channels, sample the source and not the matrix elements: 0.3 (M).

A.2 Equal variances do not mean equal disorder 1.2 pt

(a) $\overline{\epsilon_j} = f_s\epsilon_s$; $\kappa_2 = f_s(1-f_s)\epsilon_s^2$;

$$\kappa_3 = (1-f_s)(-f_s\epsilon_s)^3 + f_s\big((1-f_s)\epsilon_s\big)^3 = f_s(1-f_s)\big[(1-f_s)^2 - f_s^2\big]\epsilon_s^3 = \boxed{f_s(1-f_s)(1-2f_s)\epsilon_s^3}.$$

The mean shifts all orbital energies by the same $f_s\epsilon_s$, so it shifts the band rigidly. It is absorbed into the chemical potential (or into the clean Hamiltonian); the disorder model keeps only $\epsilon_j - f_s\epsilon_s$, which has zero mean.

(b) $f_s(1-f_s)\epsilon_s^2 = W^2/12$ gives $\epsilon_s = W/\sqrt{12 f_s(1-f_s)}$, i.e. $\epsilon_s/W = \boxed{0.962}$ at $f_s = 0.1$, and

$$\frac{\kappa_3}{W^3} = \frac{1-2f_s}{12^{3/2}\sqrt{f_s(1-f_s)}} = \frac{0.8}{41.57\times0.3} = \boxed{0.064}.$$

$\kappa_3 = 0$ at $f_s = \boxed{1/2}$ (model 1 has $\kappa_3 = 0$ for every W).

(c) A third-order quantity contains averages of three random energies, for example $\overline{\epsilon_j\epsilon_l\epsilon_m}$, which is non-zero only for $j = l = m$ and equals κ_3: the electron visits the same atom three times (in a scattering amplitude, the interference of V with VG_0V). Model 1 gives zero, model 2 with $f_s = 0.1$ gives $0.064W^3$, so effects that first appear at third order, such as an asymmetry of scattering, exist only in model 2, although both models give the same second-order results (rates, broadening). Replacing ϵ_s by $-\epsilon_s$ leaves κ_2 unchanged and reverses the sign of κ_3: second-order results stay, third-order results change sign.

Marking: (a) The three cumulants: 0.3 (C); the mean is absorbed into the chemical potential: 0.2 (M). (b) ϵ_s/W, κ_3/W^3 and $f_s = 1/2$: 0.25 (C). (c) Third-order quantities need κ_3: 0.3 (M); effect of $\epsilon_s \to -\epsilon_s$: 0.15 (M).

Part B. Displaced atoms (2.2 points)

B.1 Position disorder gives correlated bonds 1.2 pt

(a) $d_j = a + u_{j+1} - u_j$, so $\delta J_j = J'(a)(u_{j+1} - u_j)$. Then $\mathrm{Var}(\delta J_j) = 2J'^2\sigma_u^2$; bonds j and $j+1$ share atom $j+1$, so $\mathrm{Cov}(\delta J_j, \delta J_{j+1}) = -J'^2\sigma_u^2$ (correlation coefficient $-1/2$); $\mathrm{Cov}(\delta J_j, \delta J_{j+n}) = 0$ for $n \geq 2$.

(b) $K_J(q) = 2J'^2\sigma_u^2 - J'^2\sigma_u^2(e^{iq} + e^{-iq}) = \boxed{2J'^2\sigma_u^2(1-\cos q) = 4J'^2\sigma_u^2\sin^2(q/2)}$, so $K_J(0) = 0$. The sum telescopes: $\sum_{j=1}^{M}\delta J_j = J'(a)(u_{M+1} - u_1)$, with variance $2J'^2\sigma_u^2$ for every M. The total length of a stretch of bonds is the distance between its two end atoms; lengthening one bond shortens its neighbours. Long-wavelength variations of the bond lengths are absent.

(c) Independent bonds: $K_J(q) = 2J'^2\sigma_u^2$ for every q (white noise) and $\mathrm{Var}\sum_{j=1}^{M}\delta J_j = 2MJ'^2\sigma_u^2$. The average hopping (the local bandwidth) of a stretch of M bonds therefore fluctuates by $\propto M^{-1/2}$ in the independent model, but only by $\propto M^{-1}$ with position disorder. Equivalently, scattering with small momentum transfer q and slow variations of the bandwidth are present in one model and absent in the other.

Marking: (a) Linearization through the shared atoms: 0.15 (M); the three covariances: 0.25 (C). (b) $K_J(q)$ and $K_J(0) = 0$: 0.15 (C); telescoping sum and its physical reason: 0.3 (M). (c) The independent model: 0.1 (C); the two scalings of the average hopping: 0.25 (M).

B.2 Hopping and interaction move together 1.0 pt

(a) $J \propto r^{-3}$ and $V_6 \propto r^{-6}$ give

$$\frac{\delta J_j}{J(a)} = -\frac{3(u_{j+1}-u_j)}{a}, \qquad \frac{\delta V_{6,j}}{V_6(a)} = -\frac{6(u_{j+1}-u_j)}{a}.$$

Both are multiples of the same variable with coefficients of the same sign: the correlation coefficient is $\boxed{+1}$. The rms values are $3\sqrt{2}\,\sigma_u/a = \boxed{0.127}$ and $6\sqrt{2}\,\sigma_u/a = \boxed{0.255}$.

(b) $\overline{J(d_j)} \approx J(a) + \frac{1}{2}J''(a)\overline{(u_{j+1}-u_j)^2}$ with $J''/J = 12/a^2$ and $V_6''/V_6 = 42/a^2$:

$$\frac{\overline{J_j}}{J(a)} - 1 = \frac{12\sigma_u^2}{a^2} = \boxed{1.1\%}, \qquad \frac{\overline{V_{6,j}}}{V_6(a)} - 1 = \frac{42\sigma_u^2}{a^2} = \boxed{3.8\%}.$$

(A Monte Carlo average of the exact functions gives 1.09% and 3.9%.) Because $J(r)$ and $V_6(r)$ are convex, disorder increases their means: the clean values are not the averages.

(c) Any two of: the perfect correlation of J_j and $V_{6,j}$ (a short bond has both a large hopping and a large interaction); the anticorrelation of neighbouring bonds and the absence of long-wavelength fluctuations (B.1); the shifted means of (b).

Marking: (a) The two relative changes: 0.2 (C); correlation $+1$: 0.15 (M); rms values: 0.1 (C). (b) Mean shifts: 0.2 (C); the convex dependence shifts the mean: 0.1 (M). (c) Two missed features: 0.25 (M).

Part C. From correlations to rates (3.2 points)

C.1 Golden-rule rate, broadening, mean free path — and their limit 1.3 pt

(a) $V_{k'k} = \frac{1}{L}\sum_j \epsilon_j e^{i(k-k')j}$, so

$$\overline{|V_{k'k}|^2} = \frac{1}{L^2}\sum_{j,l}\overline{\epsilon_j\epsilon_l}e^{i(k-k')(j-l)} = \frac{1}{L^2}\sum_j\overline{\epsilon_j^2} = \frac{W^2}{12L}.$$

(b) $\frac{1}{\tau} = \frac{2\pi}{\hbar}\sum_{k'}\frac{W^2}{12L}\delta(E - E_{k'}) = \frac{2\pi}{\hbar}\frac{W^2}{12}\nu(E)$. At $E = J$, $\nu = 1/(\pi\sqrt{3}J)$:

$$\frac{1}{\tau} = \frac{W^2}{6\sqrt{3}\,\hbar J}, \qquad \Gamma = \frac{\hbar}{2\tau} = \frac{W^2}{12\sqrt{3}\,J} = \boxed{0.048\,J}, \qquad \ell = |v_k|\tau = \sqrt{3}\frac{Ja}{\hbar}\cdot\frac{6\sqrt{3}\hbar J}{W^2} = \frac{18J^2a}{W^2} = \boxed{18\,a}$$

for $W = J$. (An exact time evolution of a plane wave on a 3000-site ring at $W = 0.5J$ gives a decay rate within 3% of this formula.)

(c) $\xi = 24 \cdot 3J^2a/W^2 = 72J^2a/W^2$, so $\xi/\ell = \boxed{4}$: the waves are localized after a few mean free paths. For $L_\mathrm{w} = 1000a \gg \xi = 72a$: $e^{-2L_\mathrm{w}/\xi} = e^{-27.8} \approx \boxed{9\times10^{-13}}$, whereas $\ell/L_\mathrm{w} = \boxed{0.018}$ (both in units of e^2/h). The τ-based estimate is wrong by ten orders of magnitude: τ contains no interference. The decisive comparison is L_w versus ξ (τ alone is meaningful only for $L_\mathrm{w} \ll \xi$). At finite temperature, if $\ell \ll L_\varphi \ll \xi$, interference is cut off before it localizes the electron, and a τ-based description holds on scales larger than L_φ.

Marking: (a) Average over independent energies: 0.2 (C). (b) Rate from the density of final states: 0.2 (M); $1/\tau$, Γ and ℓ with numbers: 0.3 (C). (c) $\xi/\ell = 4$ and the two conductances: 0.1 (C); $L_\mathrm{w} \gg \xi$, localized, τ alone meaningless, the decisive comparison: 0.3 (M); condition $\ell \ll L_\varphi \ll \xi$: 0.2 (M).

C.2 Same broadening, different mobility **1.1 pt**

(a) For $\lambda \to 0$, K_U is constant; $\int_{-\pi}^{\pi} \mathrm{d}\theta = \int_{-\pi}^{\pi}(1-\cos\theta)\mathrm{d}\theta = 2\pi$, so $\tau_\mathrm{tr}/\tau_q = \boxed{1}$.

(b) For $k_\mathrm{F}\lambda \gg 1$ only small angles contribute: $Q^2 \approx k_\mathrm{F}^2\theta^2$, $K_U \propto e^{-\alpha\theta^2/2}$ with $\alpha = (k_\mathrm{F}\lambda)^2$, $1-\cos\theta \approx \theta^2/2$, and the limits can be extended to $\pm\infty$:

$$\frac{\tau_\mathrm{tr}}{\tau_q} = \frac{\int e^{-\alpha\theta^2/2}\mathrm{d}\theta}{\int e^{-\alpha\theta^2/2}\,\frac{\theta^2}{2}\,\mathrm{d}\theta} = \frac{\sqrt{2\pi/\alpha}}{\frac{1}{2}\sqrt{2\pi/\alpha}/\alpha} = 2\alpha = 2(k_\mathrm{F}\lambda)^2.$$

(c) At equal τ_q, the mobility ratio is the ratio of τ_tr: $2\times3^2 = \boxed{18}$ (the exact angular integral gives 17.4), in favour of the smooth disorder. A broadening Γ added to a clean formula captures the lifetime of a momentum state (the width of a spectral line). It misses the angular distribution of the scattering, which decides how fast the current direction is lost (the transport time; in diagrammatic language, vertex corrections), and every effect beyond second order in the random potential.

Marking: (a) 0.15 (C). (b) Small-angle reduction: 0.2 (M); Gaussian integrals: 0.2 (C). (c) Ratio 18: 0.15 (C); what Γ captures and what it misses: 0.4 (M).

C.3 Static disorder alone does not make a decay **0.8 pt**

(a) With $\Delta = \epsilon_1 - \epsilon_2$,

$$P_2(t) = \frac{4J^2}{\Delta^2+4J^2}\sin^2\!\Big(\frac{\sqrt{\Delta^2+4J^2}}{2\hbar}t\Big).$$

It is bounded and periodic: the detuning reduces and speeds up the oscillation, but nothing leaves irreversibly. Static disorder is a unitary, time-independent part of H.

(b) (i) A decay at rate $1/\tau$ is seen only for $\tau \ll t \ll t_\mathrm{H}$; this needs $\tau \ll t_\mathrm{H}$, equivalently many levels within the width Γ. For later times the discrete spectrum gives recurrences. (ii) $\Gamma \ll 4J$: the width must be small compared with the band width, over which $\nu(E)$ and the matrix elements change (weak disorder, $\ell \gg a$), so that second order is enough. At $E = J$, $L = 1000$:

$$t_\mathrm{H} = 2\pi\hbar L\nu(J) = \frac{2000}{\sqrt{3}}\frac{\hbar}{J} = \boxed{1155\,\hbar/J} \gg \tau = 6\sqrt{3}\,\hbar/J = 10.4\,\hbar/J.$$

Marking: (a) $P_2(t)$: 0.2 (C); no irreversible decay: 0.15 (M). (b) The two conditions: 0.3 (M); t_H and comparison: 0.15 (C).

Part D. Which random term first? (2.2 points)

D.1 Size is not importance: spectrum and observation time 1.0 pt

(a) $S(\omega_0) \propto \tau_c/(1+\omega_0^2\tau_c^2)$ is maximal at $\tau_c = 1/\omega_0 = \hbar/(10\ \text{meV}) = 66$ fs, where $S = \sigma_\eta^2/\omega_0$:

$$r_{10}^{\max} = \frac{|A_{10}|^2\sigma_\eta^2}{\hbar^2\omega_0} = \frac{(0.1\ \text{meV})^2}{\hbar\cdot 10\ \text{meV}} = \frac{10^{-3}\ \text{meV}}{\hbar} = \boxed{1.5\times10^9\ \text{s}^{-1}}.$$

(b) $r_{10} = r_{10}^{\max}\,2\omega_0\tau_c/(1+\omega_0^2\tau_c^2)$ with $\omega_0 = 1.52\times10^{13}$ s^{-1}: $\tau_c = 1$ ns gives $\boxed{2.0\times10^5\ \text{s}^{-1}}$; $\tau_c = 1$ fs gives $\boxed{4.6\times10^7\ \text{s}^{-1}}$.

(c) $t_{\text{obs}} = 1$ ps: 1.5×10^{-3}, 2×10^{-7}, 4.6×10^{-5} — all negligible. $t_{\text{obs}} = 10\ \mu$s: 1.5×10^4, 2.0, 4.6×10^2 — all three matter. The same noise is irrelevant for an ultrafast measurement and decisive for a slow pump: the ranking depends on the observation time, not only on σ_η. For $\tau_c = 1$ ns and $t_{\text{obs}} = 1$ ps the noise is frozen during the measurement: it acts as a static random shift (part (d)), not as a rate.

(d) For $\tau_c \to \infty$, $S(\omega_0) \approx 2\sigma_\eta^2/(\omega_0^2\tau_c) \to 0$: all the noise power sits at $\omega = 0$ and causes no transitions between non-degenerate levels. Static disorder belongs in H_C: solve (diagonalize, or expand) the Hamiltonian including it, compute the observable in each configuration, then average.

(e) The eigenstates have a definite electron number; $|0\rangle, |1\rangle$ belong to the same N_0, so $\langle 1|\hat{N}|0\rangle = N_0\langle 1|0\rangle = 0$ and $r_{10} = \boxed{0}$ for any σ_η: the noise only adds a global phase. (If the sample exchanges electrons with leads, the same noise shifts its potential relative to them and does matter.)

Marking: (a) Optimal τ_c and maximal rate: 0.2 (C). (b) The two rates: 0.15 (C). (c) The products: 0.1 (C); the importance depends on t_{obs}, and frozen noise for $\tau_c \gg t_{\text{obs}}$: 0.2 (M). (d) No transitions; static disorder goes into H_C: 0.2 (M). (e) Zero, and why: 0.15 (M).

D.2 A mean shift survives averaging; the spread does not 0.8 pt

(a) $\chi''(\mu_0) = -2\chi_0/\Lambda^2$, so $\overline{\chi}/\chi_0 - 1 \approx -\sigma_\mu^2/\Lambda^2 = \boxed{-9\%}$ (the exact Gaussian average gives -7.3%). At $\mu_0 + \Lambda$: $\chi = \chi_0/2$ and $\chi' = -\chi_0/(2\Lambda)$, so the rms local variation is $\chi_0\sigma_\mu/(2\Lambda)$, i.e. $\sigma_\mu/\Lambda = \boxed{30\%}$ of $\chi(\mu_0+\Lambda)$.

(b) The mean shift survives: it is a bias of every puddle in the same direction. The variation of the spatial average over N_p independent puddles decreases as $N_p^{-1/2}$ (self-averaging).

(c) Any two of: $\sigma_\mu \gtrsim \Lambda$ (the Taylor expansion fails; average the full function); a non-local response (puddles smaller than the mean free path or coherence length); non-uniform current paths and local fields (then solve $\nabla\cdot\boldsymbol{j} = 0$ with the local coefficients, as in an effective-medium or resistor-network model); puddles comparable to the sample size (no self-averaging).

Marking: (a) Mean shift: 0.15 (C); relative spread at the flank: 0.15 (C). (b) Which survives, and $N_p^{-1/2}$: 0.25 (M). (c) Two failure conditions: 0.25 (M).

D.3 Three tests 0.4 pt

1. *Coupling and symmetry:* compare the matrix element of the random term between the states that decide the observable with their energy separation, $|\langle n|\delta H|m\rangle|/|E_n - E_m|$, and check whether it breaks a symmetry the observable needs (A.1, D.1(e)).
2. *Time scale:* compare its spectral weight at the relevant transition frequencies, times the observation time, with one ($r\,t_{\text{obs}}$); a static term (all weight at $\omega = 0$) goes into H_C (D.1).
3. *Effect on the observable:* compare its mean shift and its sample-to-sample spread with the required accuracy, and check whether one rate or broadening is enough, or the angular structure, higher cumulants and interference are needed (A.2, C.1, C.2, D.2).

Marking: Coupling test with its comparison: 0.15 (M). Time-scale test: 0.15 (M). Effect on the observable: 0.1 (M).

Problem 2: Impurities and the second-order Hall current (10.0 points)

Part A. Three sources of a Hall current (2.2 points)

A.1 Intrinsic Hall conductivity of one cone 0.9 pt

(a) $\boldsymbol{d} = (vk_x, vk_y, m)$, $\partial_{k_x}\boldsymbol{d} \times \partial_{k_y}\boldsymbol{d} = (0, 0, v^2)$, $\boldsymbol{d} \cdot (\dots) = mv^2$:

$$\Omega_\pm(\boldsymbol{k}) = \mp\frac{mv^2}{2\varepsilon_{\boldsymbol{k}}^3} = \mp\frac{mv^2}{2(v^2k^2 + m^2)^{3/2}}.$$

(b) The anomalous velocity is $-\dot{\boldsymbol{k}} \times \Omega\hat{\boldsymbol{z}} = (e/\hbar)\boldsymbol{E} \times \Omega\hat{\boldsymbol{z}}$; for $\boldsymbol{E} = E_x\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{x}} \times \hat{\boldsymbol{z}} = -\hat{\boldsymbol{y}}$, so $\dot{y}_{\mathrm{an}} = -(e/\hbar)E_x\Omega$ and $j_y = -e\sum_n \int f_{0,n}\dot{y}_{\mathrm{an}} = (e^2/\hbar)E_x \sum_n \int f_{0,n}\Omega_n$. With $\int \frac{\mathrm{d}^2k}{(2\pi)^2} = \frac{1}{2\pi}\int k\,\mathrm{d}k$ and $\int_0^{k_1} \frac{v^2 k\,\mathrm{d}k}{\varepsilon_{\boldsymbol{k}}^3} = \frac{1}{m} - \frac{1}{\varepsilon_{k_1}}$:

$$\text{lower band (all } k\text{)}: \ \frac{m}{4\pi}\cdot\frac{1}{m} = \frac{1}{4\pi}; \qquad \text{upper band } (k < k_{\mathrm{F}}): \ -\frac{m}{4\pi}\Big(\frac{1}{m} - \frac{1}{\varepsilon_{\mathrm{F}}}\Big).$$

The sum is $m/(4\pi\varepsilon_{\mathrm{F}})$, so

$$\sigma_{yx} = \frac{e^2}{\hbar}\frac{m}{4\pi\varepsilon_{\mathrm{F}}} = \boxed{\frac{e^2}{h}\frac{m}{2\varepsilon_{\mathrm{F}}}}.$$

The filled lower band gives half a conductance quantum; the occupied part of the upper band reduces it.

Marking: (a) 0.15 (C). (b) The anomalous current summed over the occupied states of both bands: 0.3 (M); the two band integrals: 0.25 (C); result in e^2/h: 0.1 (C); reading of the two band contributions: 0.1 (M).

A.2 Time reversal cancels the linear Hall current 0.5 pt

With $\boldsymbol{k} \to -\boldsymbol{k}$, $\int \tilde{f}_0\tilde{\Omega}\,\mathrm{d}^2k = -\int f_0(-\boldsymbol{k})\Omega(-\boldsymbol{k})\,\mathrm{d}^2k = -\int f_0\Omega\,\mathrm{d}^2k$: the partner gives $-\sigma_{yx}$ and the total vanishes. The Berry curvature is odd under time reversal; any Hall conductivity linear in $\boldsymbol{E}$ is odd under time reversal and therefore vanishes in a time-reversal-symmetric crystal, whatever its microscopic source.

Marking: Change of variables: 0.2 (C). The general statement: 0.3 (M).

A.3 Three of four linear mechanisms scale alike 0.8 pt

Because $1/\tau \propto n_i$, each factor n_i counts as τ^{-1}; the linear distribution change is $\delta f \propto \tau E$.

intrinsic	$f_0\times$ anomalous velocity	τ^0
side jump	$\delta f \times v^{\mathrm{sj}} \propto \tau\cdot\tau^{-1}$	τ^0
extrinsic skew	$g \propto \tau\, w_A^{(3)}\delta f \propto \tau\cdot n_i\cdot\tau$	τ^1
intrinsic skew	$g \propto \tau\cdot n_i^2\cdot\tau$	τ^0

Intrinsic, side-jump and intrinsic-skew contributions all scale as τ^0: changing n_i cannot separate them. Only extrinsic skew scattering ($\propto \tau \propto \sigma_{xx}$) is separable; this is the origin of fits of the form $\sigma_{\mathrm{AH}} = a\,\sigma_{xx} + b$.

Marking: Counting impurity factors as powers of τ: 0.2 (M). The four powers: 0.3 (C). Which cannot be separated: 0.3 (M).

Part B. A dipole of curvature (2.8 points)

B.1 The curvature-dipole current 0.9 pt

The Hall part at second order is f_1 times the anomalous velocity:

$$j_y = -e\int f_1\Big(-\frac{e}{\hbar}E_x\Omega\Big) = \frac{e^3\tau}{\hbar^2}E_x^2\int \Omega\,\partial_{k_x}f_0 = -\frac{e^3\tau}{\hbar^2}D_xE_x^2,$$

using integration by parts (summed over bands). For the partner, $\partial_{k_x}[\Omega(-\boldsymbol{k})] = -(\partial_{k_x}\Omega)(-\boldsymbol{k})$, so $\tilde{D}_x = \int f_0(-\boldsymbol{k})(\partial_{k_x}\Omega)(-\boldsymbol{k})\,\mathrm{d}^2k/(2\pi)^2 = D_x$: the two cones add. Time reversal makes Ω odd and f_0 even in $\boldsymbol{k}$, so $\int f_0\Omega$ cancels between partners; the field-shifted distribution $f_1 \propto \partial_{k_x} f_0$ is odd, and its product with the odd Ω is even and survives.

Marking: Second-order current: 0.3 (C). Partner adds: 0.2 (C). Only f_1 times the anomalous velocity is the Hall part: 0.1 (M). Why time reversal allows it: 0.3 (M).

B.2 Dipole of the tilted cone **1.2 pt**

Ω does not depend on v_t (the tilt multiplies the identity). For the upper band

$$f_0 = \Theta(\varepsilon_\mathrm{F} - \varepsilon_{\boldsymbol{k}} - v_t k_x) \approx \Theta(\varepsilon_\mathrm{F} - \varepsilon_{\boldsymbol{k}}) - v_t k_x\,\delta(\varepsilon_{\boldsymbol{k}} - \varepsilon_\mathrm{F}).$$

The zeroth-order term gives zero: $\Theta(\varepsilon_\mathrm{F} - \varepsilon_{\boldsymbol{k}})$ is even and $\partial_{k_x}\Omega$ odd in k_x. With $\partial_{k_x}\Omega = \Omega'(k)k_x/k$ and $\Omega'(k) = 3mv^4k/(2\varepsilon_{\boldsymbol{k}}^5)$,

$$D_x = -\frac{v_t}{(2\pi)^2}\int \mathrm{d}^2k\,k_x^2\frac{3mv^4}{2\varepsilon_{\boldsymbol{k}}^5}\delta(\varepsilon_{\boldsymbol{k}} - \varepsilon_\mathrm{F}) = -\frac{v_t}{4\pi^2}\,\pi\int_0^\infty k^3\frac{3mv^4}{2\varepsilon_{\boldsymbol{k}}^5}\frac{\varepsilon_\mathrm{F}}{v^2k_\mathrm{F}}\delta(k-k_\mathrm{F})\,\mathrm{d}k = -\frac{3v_tmv^2k_\mathrm{F}^2}{8\pi\varepsilon_\mathrm{F}^4},$$

and $v^2k_\mathrm{F}^2 = \varepsilon_\mathrm{F}^2 - m^2$ gives the stated result. The lower band is completely filled for $v_t < v$ (its maximum lies below zero), so its contribution $\int \partial_{k_x}\Omega_-\,\mathrm{d}^2k$ is the integral of a derivative that vanishes at large k. For $v_t = 0$ the Fermi sea is a disk centred on the curvature maximum: $\partial_{k_x}\Omega$ is odd and averages to zero. A dipole needs a Fermi sea displaced relative to the curvature, here provided by the tilt (a broken mirror $k_x \to -k_x$). (A numerical integration at $v_t = 0.02v$, $m = 0.4\varepsilon_\mathrm{F}$ agrees to 0.1%; inserting τ reproduces the curvature-dipole coefficient of the source paper, with its convention $j = 2\chi E^2$.)

Marking: Expansion of f_0 to first order in v_t: 0.25 (M). Zeroth order vanishes: 0.1 (C). Angular and delta-function integrals: 0.4 (C). Filled lower band gives a total derivative: 0.2 (M). Why $D_x = 0$ at $v_t = 0$ (symmetry): 0.25 (M).

B.3 At optical frequencies the dipole current forgets τ **0.7 pt**

At first order, $f_1 = \mathrm{Re}(f_{1\omega}e^{i\omega t})$ gives $i\omega f_{1\omega} - (e/\hbar)\mathcal{E}\partial_{k_x}f_0 = -f_{1\omega}/\tau$, hence the stated $f_{1\omega}$. The Hall current is $j_y(t) = (e^2/\hbar)E_x(t)\int f_1(t)\Omega$, and the product rule gives

$$j_y^{(2\omega)} = \frac{1}{2}\frac{e^2}{\hbar}\mathcal{E}\int f_{1\omega}\Omega = \boxed{-\frac{e^3}{2\hbar^2}\frac{\tau}{1+i\omega\tau}D_x\mathcal{E}^2} \xrightarrow{\omega\tau\gg1} \frac{ie^3}{2\hbar^2\omega}D_x\mathcal{E}^2.$$

For $\omega\tau \gg 1$ the amplitude no longer depends on τ (or on the impurity density) and is phase-shifted by $90°$. A "$\propto \tau$" test of the curvature-dipole current therefore has no meaning at optical frequencies; the scaling analysis of low-frequency transport does not carry over to nonlinear optics, and when $\hbar\omega$ approaches the gap, processes between the bands beyond this wave-packet picture take over.

Marking: $f_{1\omega}$: 0.15 (C). The 2ω amplitude: 0.2 (C). High-frequency limit: 0.1 (C). Consequence for τ-based identification: 0.25 (M).

Part C. Collisions enter at the same order **(2.8 points)**

C.1 Same order as the dipole **0.9 pt**

At second order $f_2 \propto \tau^2E^2$.

curvature dipole	$f_1\times$ anomalous velocity	τ^1
side jump	$f_2 \times v^\mathrm{sj} \propto \tau^2\cdot\tau^{-1}$	τ^1
extrinsic skew	$\tau\, w_A^{(3)} f_2 \propto \tau\cdot n_i\cdot\tau^2$	τ^2
intrinsic skew	$\tau\cdot n_i^2\cdot\tau^2$	τ^1

Side jump and intrinsic skew scattering have the same n_i-dependence as the curvature dipole. A measured second-order Hall current $\propto \tau$ therefore does not identify the curvature dipole.

Marking: Second-order counting rule: 0.2 (M). The four powers: 0.3 (C). Same scaling, and $\propto \tau$ does not identify the dipole: 0.4 (M).

C.2 Equal scaling, comparable size, theory-dependent sign — 0.8 pt

(a) All three currents are $\propto \tau \propto 1/(n_i V_0^2)$ times functions of ε_F and m only; in the ratio n_i and V_0 cancel and only ε_F/m remains. At $\varepsilon_\mathrm{F} = 2m$: $|\chi^\mathrm{sj}/\chi^\mathrm{cd}| = 21/21 = \boxed{1.0}$ and $|\chi^\mathrm{isk}/\chi^\mathrm{cd}| = 3 \cdot 129/(6 \cdot 49) = \boxed{1.3}$. The collision currents are as large as the band-geometric one.

(b) Measurements that vary the impurity density (the scaling with σ_{xx}) cannot distinguish the theories: both predict the same powers of τ, so the same scaling form. A measurement that compares the sign and size of the total signal with an independent value of D_x (from the band structure) can: at $\varepsilon_\mathrm{F} = 2m$ theory I gives a total of about $-1.3\,\chi^\mathrm{cd}$ and theory II about $+3.3\,\chi^\mathrm{cd}$, and in this model the totals keep these signs for all $\varepsilon_\mathrm{F} > m$. The shape of the ε_F-dependence of the total, compared with the predicted shapes, can also distinguish them.

Marking: (a) Independence of n_i, V_0: 0.2 (M); the two magnitudes: 0.15 (C); comparable size: 0.05 (M). (b) Scaling cannot: 0.2 (M); what can: 0.2 (M).

C.3 The scaling law and its intercept — 1.1 pt

(a) No current along y: $0 = \sigma E_y + \chi E_x^2$, so

$$\frac{E_y}{E_x^2} = -\frac{A_1\tau + A_2\tau^2}{\beta\tau} = -\frac{A_2}{\beta^2}\,\sigma - \frac{A_1}{\beta}, \qquad C_1 = -\frac{A_2}{\beta^2}, \quad C_2 = -\frac{A_1}{\beta}.$$

(b) The three points lie on one line: $C_1 = \boxed{0.4 \times 10^{-6}\ \mathrm{m\,V^{-1}mS^{-1}}} = 4 \times 10^{-4}$ m/(V S), and $C_2 = \boxed{0.5 \times 10^{-6}\ \mathrm{m/V}}$.

(c) C_1: the τ^2 mechanism, extrinsic skew scattering ($\propto V_1^3$). C_2: all τ^1 mechanisms — curvature dipole, side jump and intrinsic skew. C_2 does not measure the curvature dipole alone.

(d) Phonons add a second scattering source, $\rho = \rho_0 + \rho_T$; each source has its own side-jump and skew coefficients, mixed terms $\propto \rho_0\rho_T$ appear, and the time-dependent phonon field gives no extrinsic-skew term of the static kind. The relation becomes $E_y/E_x^2 = (C_1'\rho_0 + C_2'\rho_0^2 + C_3'\rho_0\rho_T + C_4'\rho_T^2)/\rho^2$, which is not one straight line in σ (it reduces to (a) only when one source dominates).

Marking: (a) Open-circuit condition: 0.2 (M); C_1, C_2: 0.15 (C). (b) Numbers with units: 0.15 (C). (c) Assignment of mechanisms: 0.15 (M); C_2 is not the dipole: 0.15 (M). (d) Two sources and their mixing: 0.3 (M).

Part D. One broadening, many ensembles — (2.2 points)

D.1 One Γ, three different responses — 0.8 pt

(a) $\overline{V_i^2} = V_0^2$ in all three; $\overline{V_i^3} = +V_0^3, -V_0^3, 0$.

(b) The same: $1/\tau$, Γ (both $\propto n_i V_0^2$), χ^cd ($\propto \tau$), χ^sj and χ^isk (they depend only on $n_i V_0^2$ and the bands). Different: $\chi^\mathrm{esk} \propto V_1^3$, which is positive, negative and zero. (The mean potential $\pm n_i V_0$ only shifts the chemical potential.)

(c) No. Γ is fixed by the second moment; the second-order response also depends on the third moment of the disorder (and on its correlations). A theory that only adds $i\Gamma$ to clean formulas cannot tell the three ensembles apart.

Marking: (a) 0.15 (C). (b) Classification: 0.35 (M). (c) Γ is not enough: 0.3 (M).

D.2 Average inversion symmetry: protects edges, forbids the bulk response **0.9 pt**

(a) Apply inversion to the whole situation: configuration C in field $\boldsymbol{E}$ becomes configuration $\mathcal{I}C$ in field $-\boldsymbol{E}$, and the current, a polar vector, becomes $-\boldsymbol{j}$; hence $\boldsymbol{j}_{\mathcal{I}C}(-\boldsymbol{E}) = -\boldsymbol{j}_C(\boldsymbol{E})$. Relabelling $C \to \mathcal{I}C$ (a one-to-one map with $P[\mathcal{I}C] = P[C]$):

$$\overline{\boldsymbol{j}}(-\boldsymbol{E}) = \sum_C P[\mathcal{I}C]\, \boldsymbol{j}_{\mathcal{I}C}(-\boldsymbol{E}) = -\sum_C P[C]\, \boldsymbol{j}_C(\boldsymbol{E}) = -\overline{\boldsymbol{j}}(\boldsymbol{E}),$$

so $\overline{\boldsymbol{j}}$ is odd in $\boldsymbol{E}$ and every even order vanishes: $\overline{\chi} = 0$. A typical configuration lacks inversion, so $\chi_C \neq 0$, with $\chi_{\mathcal{I}C} = -\chi_C$.

(b) $\chi = \frac{1}{M}\sum_i \chi_i$ with zero-mean, independent χ_i: its mean is zero and its rms decreases as $M^{-1/2}$ (self-averaging).

(c) Inversion takes the left end of C to the right end of $\mathcal{I}C$: $\epsilon_L(\mathcal{I}C) = \epsilon_R(C)$. Hence $P(\epsilon_L = E) = \sum_C P[\mathcal{I}C]\,\delta\big(E - \epsilon_L(\mathcal{I}C)\big) = \sum_C P[C]\,\delta\big(E - \epsilon_R(C)\big) = P(\epsilon_R = E)$: the edges are degenerate *on average* (statistically). Both (a) and (c) follow from one property, the invariance of the ensemble (its probability distribution over configurations) under inversion: averaged quantities that inversion exchanges are equally distributed, and averaged quantities that inversion reverses vanish. The second-order current is reversed, so the averaged bulk response vanishes, while a phase distinction defined for the ensemble can survive.

Marking: (a) Symmetry relation: 0.2 (M); average vanishes: 0.15 (C); typical $\chi_C \neq 0$: 0.05 (M). (b) $M^{-1/2}$: 0.15 (M). (c) Equal distributions: 0.15 (C); same principle: 0.2 (M).

D.3 What the scaling data can and cannot tell **0.5 pt**

Determined: C_1 (extrinsic skew, $\propto V_1^3$), C_2 (sum of all τ^1 terms), and the fact that the ensemble breaks inversion on average (otherwise the averaged signal vanishes). *Not determined:* how C_2 splits into curvature dipole, side jump and intrinsic skew — they have the same n_i-dependence. *Inputs needed* (any two of): (1) the factor $\beta = \sigma/\tau$ (from the carrier density and mass, or τ from an independent line width), which converts C_2 into the coefficient A_1; (2) an independent value of D_x from the band structure; (3) a microscopic model of the actual disorder (its moments and correlations) with a theory of the τ^1 collision coefficients at a trusted level — C.2(b) shows that the level of theory can change their sign.

Marking: What is determined: 0.15 (M). What is not: 0.15 (M). Two of the three inputs: 0.2 (M).

Problem 3: Disorder in a two-cycle quantum pump (10.0 points)

Part A. Topology from disorder in a dimer chain (1.6 points)

A.1 The edge state is a product of random ratios 0.8 pt

(a) With $\psi(b_n) = 0$, $E = 0$ requires $(H_\mathrm{d}\psi)(b_n) = u_n\psi(a_n) + v_n\psi(a_{n+1}) = 0$, i.e. $\psi(a_{n+1}) = -(u_n/v_n)\psi(a_n)$; the equations at the a sites hold automatically. (On a finite chain ending with b_N, the last equation $u_N\psi(a_N) = 0$ fails only by the exponentially small amplitude at the right end.)

(b) $\psi(a_n) = \psi(a_1)\prod_{m=1}^{n-1}(-u_m/v_m)$, so $\ln|\psi(a_n)/\psi(a_1)| = \sum_{m<n}\ln|u_m/v_m| \approx n\,\overline{\ln|u/v|}$. A left edge state exists if and only if

$$\overline{\ln|u|} < \overline{\ln|v|}, \qquad \text{decay length } = \frac{1}{\overline{\ln|v|} - \overline{\ln|u|}} \text{ cells.}$$

For a regular chain this is $|v| > |u|$.

Marking: (a) 0.2 (C). (b) Product and law of large numbers: 0.15 (C); condition: 0.25 (M); decay length: 0.1 (C); the regular criterion as a special case: 0.1 (M).

A.2 Disorder makes the chain topological 0.8 pt

(a) $W = 0$: $u = 1 > v = 0.8$, trivial, no edge state. $\overline{\ln u} = \frac{1}{W}\left[x\ln x - x\right]_{1-W/2}^{1+W/2}$:

$$W = 1: \quad \overline{\ln u} = (1.5\ln 1.5 - 1.5) - (0.5\ln 0.5 - 0.5) = \boxed{-0.045};$$
$$W = 2: \quad \overline{\ln u} = \tfrac{1}{2}(2\ln 2 - 2) = \ln 2 - 1 = \boxed{-0.307}.$$

With $\ln 0.8 = -0.223$: no edge state at $W = 1$ ($-0.045 > -0.223$), an edge state at $W = 2$ ($-0.307 < -0.223$), with decay length $1/(0.307 - 0.223) = \boxed{12 \text{ cells}}$. (The transition lies at $W \approx 1.87$.)

(b) The edge-state amplitude is a product of random ratios; its logarithm is a sum, whose typical value is $n\overline{\ln|u/v|}$ — the geometric mean decides. The arithmetic mean $\overline{u} = 1 > 0.8$ would predict "trivial" for every W. Because $\ln$ is concave, $\overline{\ln u} < \ln\overline{u}$, and rare weak intracell bonds ($u \to 0$, $\ln u \to -\infty$) carry a large weight: they cut the chain into pieces bound across the cells. Disorder alone drives the chain from trivial to topological.

Marking: (a) Phase at $W = 0$: 0.05 (M); the two logarithmic means: 0.2 (C); the two decisions: 0.1 (M); decay length: 0.05 (C). (b) Why the arithmetic mean fails and the logarithmic mean decides: 0.4 (M).

Part B. Counting the pumped charge (2.0 points)

B.1 The current operator and local bookkeeping 1.0 pt

(a) $\dot{n}_j = i[H, n_j]$. The on-site and density terms commute with n_j and drop out. For the hopping, $[c_{l+1}^\dagger c_l, n_j] = c_{l+1}^\dagger c_l(\delta_{lj} - \delta_{l+1,j})$ gives

$$\dot{n}_j = i\left(J_{j-1}c_j^\dagger c_{j-1} - J_{j-1}^* c_{j-1}^\dagger c_j\right) - i\left(J_j c_{j+1}^\dagger c_j - J_j^* c_j^\dagger c_{j+1}\right) = I_{j-1} - I_j.$$

The random energies change the state, not the current operator. (Random hoppings would enter I_j.)

(b) $\frac{\mathrm{d}}{\mathrm{d}t}\sum_{j=j_1}^{j_2} n_j = \sum_{j=j_1}^{j_2}(I_{j-1} - I_j) = I_{j_1-1} - I_{j_2}$; integrate from t_i to t_f: $\Delta N = Q_{j_1-1} - Q_{j_2}$.

(c) Sites 11–30: $Q_{10} - Q_{30} = -0.01 \approx 0$. Sites 31–50: $Q_{30} - Q_{50} = \boxed{-0.37}$ (depleted). Sites 51–70: $Q_{50} - Q_{70} = \boxed{+0.39}$ (filled up). Charge has been moved from the region left of bond 50 to the region right of it: a local rearrangement (for example, particles held in a region where the pump has stopped), not uniform transport. One number per period does not summarize this window: in uniform periodic transport the density returns after each window and all bonds carry the same charge. One must report the charge at several bonds and over several windows.

Marking: (a) Commutator and continuity equation: 0.25 (C); on-site and density terms drop out, meaning: 0.15 (M). (b) 0.1 (C). (c) The three changes: 0.1 (C); local rearrangement: 0.2 (M); one number is not a valid summary: 0.2 (M).

B.2 A Chern number without momentum, and its fraction **1.0 pt**

(a) The definition uses only the Hamiltonian of one ring, a flux through it and the ground state: no translation symmetry, no Bloch momentum. The boundary phase θ plays the role of the momentum, and the average over θ replaces the integral over k. Each configuration has its own Chern number; while the gap stays open it is an integer that cannot change continuously.

(b) Starting in $|\Psi_A\rangle$, the first period carries q_1 and ends in $|\Psi_B\rangle$; since H has period T, the second period carries q_2. Over two periods $q_1 + q_2 = C$, so $\boxed{C/2}$ per period; the same holds starting in $|\Psi_B\rangle$. The single-period values need not be equal or quantized. The data give $-0.582 - 0.418 = -1.000$: consistent with $C = -1$, i.e. $-1/2$ per period on average.

(c) The gap that separates the pair $\{|\Psi_A\rangle, |\Psi_B\rangle\}$ from all other states must stay open uniformly in the chain length (for all t and θ).

Marking: (a) 0.3 (M). (b) Two-period argument: 0.2 (C); $C/2$ and unequal single periods: 0.1 (M); data consistent: 0.1 (C). (c) Uniform gap of the pair: 0.3 (M).

Part C. What a random potential does to the flavors **(2.6 points)**

C.1 Bias and mixing from one random potential **1.2 pt**

(a) $h_z = \frac{1}{2}(M_{AA} - M_{BB}) = \frac{1}{2}\sum_j \epsilon_j\, d_R(j)$ and $h_x - ih_y = M_{AB} = \sum_j \epsilon_j\, w^*_{R,A}(j) w_{R,B}(j)$.

(b) $\overline{h_z} = \frac{1}{2}\sum_j \overline{\epsilon_j} d_R(j) = 0$; $\mathrm{Var}\, h_z = \frac{1}{4}\sum_{j,l} \overline{\epsilon_j \epsilon_l} d_R(j) d_R(l) = \frac{1}{4}\frac{W^2}{12}\sum_j d_R(j)^2 = \frac{W^2}{48}\sum_j d_R(j)^2$, and the covariance follows in the same way. Being sums of independent site energies, the biases of two cells are independent when d_R and $d_{R'}$ have no site in common.

(c) $M_{AA} = \epsilon_1 \cos^2\phi + \epsilon_2 \sin^2\phi$, $M_{BB} = \epsilon_1 \sin^2\phi + \epsilon_2 \cos^2\phi$, $M_{AB} = (\epsilon_2 - \epsilon_1)\sin\phi\cos\phi$:

$$h_z = \tfrac{1}{2}(\epsilon_1 - \epsilon_2)\cos 2\phi, \qquad h_x = -\tfrac{1}{2}(\epsilon_1 - \epsilon_2)\sin 2\phi, \qquad h_y = 0, \qquad h_x^2 + h_z^2 = \tfrac{1}{4}(\epsilon_1 - \epsilon_2)^2.$$

The random potential fixes the size of the effective field; the shapes of the two orbitals fix its direction. Orbitals on different sites ($\phi = 0$) feel a pure bias; orbitals spread equally over both sites with opposite relative sign ($\phi = \pi/4$, bonding and antibonding) feel a pure mixing.

Marking: (a) 0.2 (C). (b) Mean, variance and covariance: 0.25 (C); independence condition: 0.15 (M). (c) h_z and h_x: 0.15 (C); invariance: 0.05 (C); what decides bias versus mixing: 0.4 (M).

C.2 For the pump's flavors, disorder is almost a pure bias **0.8 pt**

(a) $d_R = (-0.47, -0.50, 0, +0.50, +0.47)$ on $(R,1), \ldots, (R{+}1,0)$: $\sum_j d_R(j)^2 = \boxed{0.94}$, rms $h_z = W\sqrt{0.94/48} = \boxed{0.14\,W}$.

(b) $\mathrm{Var}\, h_x = \frac{W^2}{12}(0.013^2 + 0.027^2 + 0.013^2) = \frac{W^2}{12} \times 1.07 \times 10^{-3}$: rms $h_x = \boxed{0.0094\,W}$. The bias dominates by a factor of about 15: these two orbitals hardly share sites, so the random potential acts almost purely as a random flavor bias. (A scan of the full single-particle model over the cycle gives rms bias 0.135–$0.140\,W$ and rms mixing 0.008–$0.016\,W$.)

(c) The orbitals of cell R occupy $(R,1), \ldots, (R{+}1,0)$ and those of cell $R+1$ occupy $(R{+}1,1), \ldots, (R{+}2,0)$: without the neglected tails they share no site, so the biases of neighbouring cells are uncorrelated (independent, $\sigma_h \approx 0.14W$).

Marking: (a) $\sum d^2$ and rms bias: 0.2 (C). (b) rms mixing: 0.15 (C); the bias dominates, and why: 0.2 (M). (c) No shared sites, independent biases: 0.25 (M).

C.3 A static potential is an alternating field for the followed flavor **0.6 pt**

At time t the particle of cell R follows A; moving it to B releases the energy $M_{AA}-M_{BB}$, so $g_R^{\mathrm{f}}(t) = 2h_z^{(R)}(t)$. At $t+T$ the orbitals and the ϵ_j are the same, so $h_z^{(R)}(t+T) = h_z^{(R)}(t)$ for orbitals labelled by position; but the particle now follows a B orbital, and moving it to A releases $M_{BB} - M_{AA} = -2h_z^{(R)}$. Hence $g_R^{\mathrm{f}}(t+T) = -g_R^{\mathrm{f}}(t)$ (the cell label may shift because the particle has moved by part of a cell). The static random potential acts on the followed flavor as a random field that reverses its sign every period. A block that is favourable to flip during one period is unfavourable during the next, for the random part of its energy: a domain energy at one instant describes at most one period, and the lifetime depends on the field along the full two-period cycle and on the dynamics within each period.

Marking: Sign relation: 0.25 (C). Consequence for instantaneous estimates: 0.35 (M).

Part D. Domains in a long disordered chain **(3.8 points)**

D.1 The clean threshold **0.6 pt**

$\Delta E = E_{\mathrm{w}} - 2\beta_{\max}\ell < 0$ requires $\ell > E_{\mathrm{w}}/(2\beta_{\max}) = 20$, so $\ell_{\min} = \boxed{21}$ cells. A long clean chain keeps its order for a long time because creating such a domain needs 21 particles to change orbital together: the rate falls exponentially with ℓ, while smaller domains cost energy. In addition, β reverses its sign every period, so a domain is favourable only during part of each two-period cycle. The order is long-lived (prethermal), not permanent.

Marking: $\ell_{\min} = 21$: 0.2 (C). Why the clean chain stays ordered for long: 0.4 (M).

D.2 Rare favourable blocks at any disorder strength **1.2 pt**

(a) $\Delta E_D < 0 \iff \sum_{R\in D} h_R > E_{\mathrm{w}}/2$; the sum is Gaussian with variance $\ell\sigma_h^2$, so

$$p(\ell) = \Phi\Big(-\frac{E_{\mathrm{w}}}{2\sigma_h\sqrt{\ell}}\Big), \qquad p = \Phi(-1) \text{ at } \ell = \Big(\frac{E_{\mathrm{w}}}{2\sigma_h}\Big)^2.$$

(b) $W = \sigma_h/0.14 = \boxed{0.36\,J}$. $p(4) = \Phi(-0.4/(2\cdot 0.05\cdot 2)) = \Phi(-2) = \boxed{0.023}$, so about $2500\times 0.023 \approx \boxed{57}$ favourable blocks of 4 cells. (The length with $p = \Phi(-1)$ is 16 cells.)

(c) For any $\sigma_h > 0$, $p(\ell) > 0$ for every ℓ and $p \to 1/2$ for $\ell \to \infty$; a chain of N cells contains about N/ℓ independent blocks, so the number of favourable blocks grows $\propto N$. The random energy of a block grows as $\sqrt{\ell}$, while the wall cost is fixed, so some block always wins (the random-field, or Imry–Ma, argument in one dimension). In the clean chain at $b = 0$, every domain costs $E_{\mathrm{w}} > 0$. In a long disordered chain the followed configuration is therefore not the lowest state at any instant, and the gap of the two-state manifold to domain states is not uniform in N: it closes in rare regions. In equilibrium the chain would break into domains of about $(E_{\mathrm{w}}/\sigma_h)^2$ cells.

Marking: (a) Condition and $p(\ell)$: 0.25 (M); the length: 0.1 (C). (b) W: 0.1 (C); $p(4)$ and the count: 0.2 (C). (c) Argument for favourable blocks at any σ_h: 0.3 (M); gap not uniform in N: 0.25 (M).

D.3 First defect versus macroscopic failure **1.0 pt**

(a) The total rate is $N\gamma$: $1 - e^{-N\gamma t_{\mathrm{first}}} = 1 - 1/e$ gives $t_{\mathrm{first}} = 1/(N\gamma)$. After a time t there are about $N\gamma t$ defects covering $N\gamma t\xi$ cells, so the charge per period has dropped by the fraction $\gamma\xi t$, and $t_\kappa = \kappa/(\gamma\xi)$.

(b) $t_{\mathrm{first}} = 1/(10^4\cdot 10^{-6}) = \boxed{100}$ periods; $t_\kappa = 0.01/(10^{-6}\cdot 10) = \boxed{1000}$ periods.

(c) $t_{\mathrm{first}} \propto 1/N$ shrinks with the system size: in a large system the first defect appears early. t_κ does not depend on N; it is the lifetime of the transport measured on the whole ring. If defects spread, a defect born at t' covers $\xi + 2s_{\mathrm{d}}(t-t')$ cells, the stopped fraction is $\gamma\xi t + \gamma s_{\mathrm{d}} t^2$, and for $s_{\mathrm{d}}t \gg \xi$, $t_\kappa \approx \sqrt{\kappa/(\gamma s_{\mathrm{d}})}$: still independent of N, now set by the growth.

Marking: (a) t_{first}: 0.15 (C); t_κ from the stopped fraction: 0.2 (M). (b) 0.1 (C). (c) Which time depends on N, and which is the lifetime: 0.3 (M); spreading: 0.1 (C); its meaning: 0.15 (M).

D.4 The missing inputs are the research **1.0 pt**

Four inputs:

1. the time-dependent random fields and mixings along the full cycle, for the actual disorder (on-site and bond) and for the interacting, correlated flavor states — not only the clean single-particle orbitals at one instant;
2. the rate of creating reversed domains in rare regions, which needs the flavor-mixing matrix elements, the drive speed and the sign reversal of the field (a rate, not an energy);
3. the motion of domain walls in the random, alternating field: pinning versus spreading (s_{d}), and whether a nucleated domain survives the next period;
4. the conversion of domains and defects into charge error (charge carried at walls, heating), together with the prepared initial state.

The Chern number of B.2 assumes a gap of the two-state manifold that is uniform in N; disorder removes it (D.2(c)), and the invariant is an adiabatic statement that contains no rate. The domain energy at one instant is not a rate: a favourable domain need not form in an isolated system, it must be created through mixing, and the random part of ΔE_D reverses its sign every period (C.3).

Marking: Four inputs: 0.6 (M). Why not the invariant: 0.2 (M). Why not the instantaneous domain energy: 0.2 (M).

Appendices: research context

Appendices A–D are not needed to solve or mark the exam. Appendix E is for the reader. Labels have the form problem.part.number. Sources are numbered as in Appendix D.

Appendix A. Paper map

What each source answers

Source	Question	Answer	Problems
[1] Nagaosa et al., RMP 2010	What produces the anomalous Hall current of a magnetic conductor, and how are the mechanisms told apart?	Band geometry (intrinsic), side jump and skew scattering; semiclassical, Kubo and Keldysh treatments agree when done consistently; scaling with σ_{xx} separates only the τ^1 skew part; the dominant mechanism changes with the conductivity regime.	2.A
[2] Du et al., Nat. Commun. 2019	How does disorder enter the time-reversal-symmetric nonlinear Hall effect?	At leading order: side jump and intrinsic (Gaussian) skew scale like the Berry curvature dipole (τ^1), extrinsic skew as τ^2; computed for a tilted massive Dirac cone; a scaling law $E_y/E_x^2 = C_1\sigma + C_2$, generalized to several scattering sources.	2.B–2.D
[3] Du et al., Nat. Commun. 2021	Does a quantum (diagrammatic) theory confirm [2]?	The dipole term agrees; the side-jump and skew terms have the opposite sign (non-crossing approximation); the total keeps the sign of the dipole term with a larger magnitude; the scaling law is unchanged.	2.C.2
[4] Hertkorn & Oreg, Nat. Phys. 2026; [5] Yue et al.; [6] Su et al.	Can an ensemble of disordered samples, none of which has the protecting symmetry, still be topological?	Yes, if the symmetry holds on average: [5] sees a disorder-induced average topological phase with statistically degenerate edges in a Rydberg array; [6] sees a phase distinction removed by symmetry-breaking disorder and restored by the disorder average.	2.D.2, 3.A
[7] Jürgensen et al., arXiv:2504.09338	Can repulsion make a pump carry $-1/2$ particle per period at integer filling?	Yes in small systems, through two-band flavors exchanged every period; for $N > N_c \approx \tilde{U}/(2\tilde{V}_{\max})$ domain excitations meet the polarized branch; a prethermal lifetime is expected, its theory left open. Disorder is not studied.	3.B–3.D
[8] Niu & Thouless, J. Phys. A 1984 (ref. 14 of [7])	Is the pumped charge quantized with disorder and interactions?	Yes: the twist-averaged charge per cycle is an integer if the ground state is unique and gapped over the whole cycle.	3.B.2

Problem 1 is built on standard material of disorder physics that the papers use without derivation: the Anderson model [9], the one-dimensional localization length [10], quantum versus transport lifetimes [11], noise spectra and transition rates [12], and charge puddles [13].

Language map

Paper term	Physical process or picture	Familiar equivalent (or "new")	How it is computed or measured
disorder configuration, realization	one arrangement of the imperfections	one sample of a random variable	draw positions and strengths, build H_C (1.A)
disorder average	average over samples	ensemble average	$\sum_C P[C]O_C$; numerically, many samples
Anderson model	random orbital energies on a lattice	tight-binding model with random diagonal	1.A.1, 1.C.1
correlated disorder, $K_J(q)$, $K_U(Q)$	how random values at different places or channels are related	covariance, power spectrum	Fourier transform of the covariance (1.B.1)
Gaussian / non-Gaussian disorder	whether moments beyond the variance matter	cumulants	$\kappa_3 = \overline{(x-\bar{x})^3}$ (1.A.2, 2.D.1)
Born approximation	scattering to lowest order in the random potential	Fermi's golden rule	$\overline{\lvert V_{k'k}\rvert^2}$ (1.C.1)
self-energy, broadening Γ	finite lifetime of a momentum state	Lorentzian line width	$\Gamma = \hbar/(2\tau_q)$ (1.C.1)
quantum vs transport lifetime	leaving a momentum state vs losing the current direction	$1-\cos\theta$ weighting of kinetic theory	1.C.2
vertex correction	angular structure of scattering inside a response	transport cross section	$(1-\cos\theta)$, ladder sums (1.C.2)
localization length ξ	decay length of eigenstates through interference	new: taught by 1.C.1(c)	transfer matrices, Lyapunov exponent
dephasing length L_φ	distance over which phase memory survives	coherence length	inelastic rates (1.C.1(c))
Heisenberg time	time to resolve the discrete levels	recurrence time	$2\pi\hbar/\delta$ (1.C.3)
noise spectral density	noise power at a frequency	power spectrum (Wiener–Khinchin)	Fourier transform of $\overline{\eta(t)\eta(0)}$ (1.D.1)
charge puddles	regions of different local chemical potential	inhomogeneous doping	scanning probes (1.D.2)
self-averaging	the sample average approaches the ensemble average	law of large numbers	spread $\propto M^{-1/2}$ (1.D.2, 2.D.2)
Berry curvature	momentum-space "magnetic field" of a band	new: taught by 2.A.1	$\Omega = \nabla_{\boldsymbol{k}} \times i\langle u\rvert\nabla_{\boldsymbol{k}} u\rangle$
anomalous velocity	sideways velocity from the curvature	Lorentz force in momentum space	$-\dot{\boldsymbol{k}} \times \Omega$ (2.A.1)
side jump	sideways shift of the packet in a collision	displacement in a scattering event	$\delta\boldsymbol{r} = \mathcal{A}_{k'} - \mathcal{A}_k - (\nabla_k + \nabla_{k'})\arg V_{k'k}$ (Appendix B)
skew scattering (extrinsic, intrinsic)	asymmetric scattering probability (one impurity at third order; pairs at fourth)	Mott asymmetry	antisymmetric part of the rate (2.A.3)
Berry curvature dipole	first moment of Ω over the Fermi sea	dipole moment	$D_x = \int f_0 \partial_{k_x}\Omega$ (2.B)
nonlinear Hall effect	transverse current $\propto E^2$	low-frequency second-harmonic generation	2ω transverse voltage (2.C.3)
scaling law	fit of the signal against σ_{xx} to separate powers of τ	Matthiessen-type analysis	$E_y/E_x^2 = C_1\sigma + C_2$ (2.C.3)
average symmetry, mixed state	ensemble invariant under a symmetry that single samples break	symmetric probability distribution	$P[\mathcal{I}C] = P[C]$; $\bar{\rho}$ invariant (2.D.2)

Paper term	Physical process or picture	Familiar equivalent (or "new")	How it is computed or measured
statistical degeneracy	two levels with equal distributions but different values in each sample	new: taught by 2.D.2(c)	distributions of ϵ_L, ϵ_R
average SPT	a topological phase defined for an ensemble	new: 2.D.2, 3.A (toy)	string orders or polarizations averaged over C and $\mathcal{I}C$
Thouless pump	slow cyclic drive that carries charge	adiabatic transport	$\int \langle I_j \rangle \mathrm{d}t$ per period (3.B.1)
many-body Chern number	integer charge per cycle without momentum	quantized Hall conductance	θ-averaged charge (3.B.2)
twisted boundary condition	flux through the ring	Aharonov–Bohm phase	one hopping times $e^{i\theta}$ (3.B.2)
flavor A/B	two localized orbitals per cell from the two lowest bands	basis inside a degenerate subspace	projected-position eigenvectors (Fig. 3.3)
domain, domain wall	region of the other flavor; its boundary	Ising domain	3.D
random-field (Imry–Ma) argument	random energy $\propto \sqrt{\ell}$ beats a fixed wall cost	random-field Ising model	3.D.2
rare region	atypical local disorder that changes local behaviour	Griffiths region	probability of a large local sum (3.D.2)
prethermal lifetime	how long a driven system keeps its order	metastable lifetime	time when transport leaves a tolerance (3.D.3)
topology from disorder	disorder drives trivial to topological	new: taught by 3.A	logarithmic-mean criterion

Chain of objects

Labels: S stated in a source; D derived in this booklet; O-calc open, existing theory is enough and a calculation is missing; O-theory open, the closest theory needs a hypothesis that fails here.

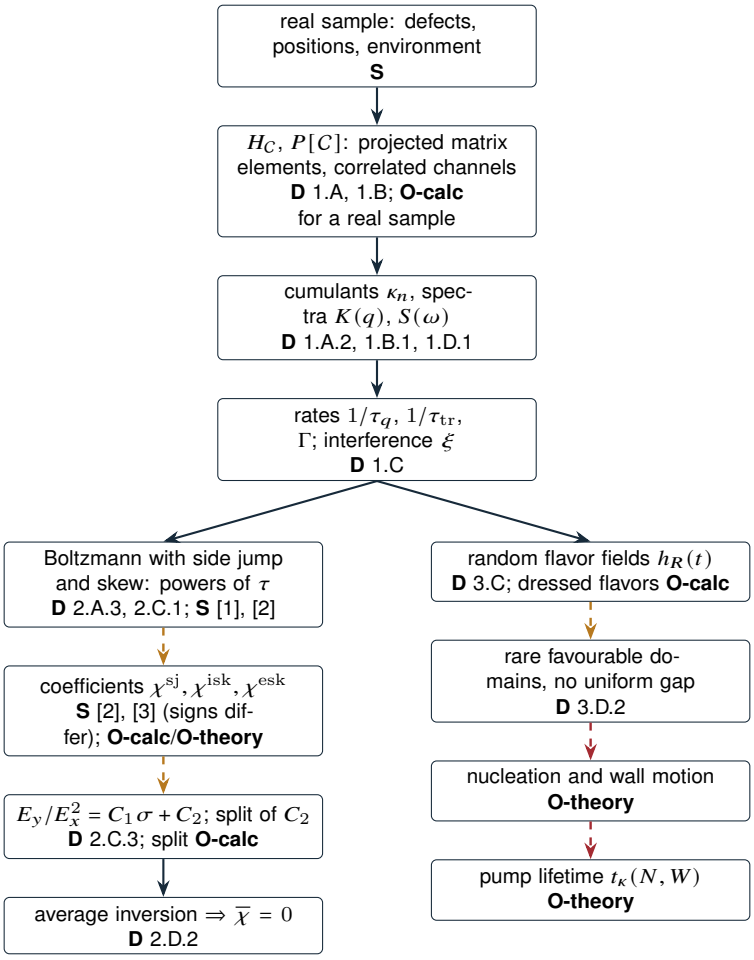


Figure A.1. From a real sample to the measured quantities. Black arrows: derived. Amber dashed: open, a calculation is missing (O-calc). Red dashed: open, new theory is needed (O-theory). A third branch (3.A: topology from disorder in a chiral chain, D; interacting long-range average SPT, S [5], [6]; which transport an average symmetry protects, O-theory) is listed in the table below.

Open links: a calculation or a new theory?

Open link (questions)	Kind	Closest existing theory	Hypothesis that fails, or the calculation that closes it
disorder model of a real sample: moments and correlations of its actual defects (1.A, 1.B, 2.D.1)	O-calc	first-principles defect potentials projected on Wannier orbitals; defect statistics from characterization	compute δh for each defect type, then κ_n and the spectra $K(q)$ for the measured concentrations
τ^1 collision coefficients for that disorder (2.C.2, 2.D.3)	O-calc	diagrammatic Kubo theory, non-crossing approximation [3]	evaluate side-jump and intrinsic-skew terms for the real bands and defect T-matrices
a semiclassical theory consistent with [3]; coefficients beyond weak disorder (2.C.2(b))	O-theory	Boltzmann theory with side jump [2]; non-crossing diagrammatics [3]	[3] finds contributions without semiclassical counterpart; near the band edge $\hbar/(\varepsilon_{\rm F}\tau)$ is not small. Holds for $\varepsilon_{\rm F}\tau/\hbar \gg 1$ with non-crossing diagrams only
collision terms at finite frequency (2.B.3)	O-calc	Kubo or Keldysh theory at finite ω	compute side jump and skew as functions of $\omega\tau$ for the model

Open link (questions)	Kind	Closest existing theory	Hypothesis that fails, or the calculation that closes it
random fields of the interacting, correlated flavors and of bond disorder over the cycle (3.C)	O-calc	Schrieffer–Wolff dressing; projection as in 3.C.1	compute $h_R(t)$ and the mixings for the many-body flavor states
nucleation of reversed domains in rare regions under a slow drive with an alternating field (3.D.2, 3.D.4)	O-theory	Landau–Zener and multiphoton rates; rigorous prethermalization theorems	they need a fast drive or a large uniform gap; here the drive is slow and rare regions have no gap
domain-wall motion: pinning versus spreading (3.D.3)	O-theory	random-field Ising wall dynamics (creep) in equilibrium; finite-time tensor-network numerics	creep theory assumes thermal activation; here isolated, periodically driven quantum walls. Holds in the strong-pinning limit
Chern number of a two-state manifold in a long disordered ring (3.B.2)	D/O-theory	Niu–Thouless(–Wu) [8]	a gap uniform in N, which disorder removes (3.D.2(c)); holds for N below the rare-region spacing
transport protected by an average symmetry (2.D.2, 3.A)	O-theory	average-SPT classification of mixed states [14]	the classification concerns ensembles of states, not their driven response; average inversion forbids odd bulk responses

Shared gap: in all O-theory links the useful quantity is not an averaged energy or a single broadening but a rate set by rare regions or a constraint set by an ensemble symmetry, and no gap or small parameter is uniform in the system size; a theory of disorder-averaged *dynamics* at this level is missing.

Appendix B. Source, scope and open conditions for each question

Problem 1

1.A.1–1.A.2. Teaching models; the projection $\delta h_{ij} = \langle \varphi_i | v | \varphi_j \rangle$ is the general recipe of tight-binding disorder models. Overlap of the Gaussian orbitals is neglected; with orthogonalized orbitals the numbers change, not the conclusion. Third-order effects: the extrinsic skew term of [1], [2] needs κ_3 (there $\overline{V_i^3} = V_1^3$).

1.B.1–1.B.2. The tweezer model follows [5], where whole cells are displaced so that the intracell distance is fixed and the long-range dipolar exchange $J_{ij} = C_3(1 - 3\cos^2\theta_{ij})/(2r_{ij}^3)$ and the van der Waals term both change. The nearest-neighbour, $\theta = 90^\circ$ version here is a reduction, not their model.

1.C.1–1.C.3. Weak-disorder Born rates; $\xi = 24(4J^2 - E^2)a/W^2$ is the perturbative one-dimensional result (it fails near $E = 0$, where the exact value is about 10% larger). The checks of Appendix D confirm both formulas at $W = 0.5J$.

NOT ESTABLISHED — O-calc — what 1.C does not establish

For a real material, the spectra $K(q)$ and the vertex structure must come from its actual defects; a single Γ fitted to a line width is not a disorder model.

1.D.1–1.D.3. Classical noise only; quantum baths need ordered correlation functions with $S(\omega) \neq S(-\omega)$ (absorption versus emission) and the bath's own occupation. The puddle estimate assumes local response and uniform fields.

Problem 2

2.A. Standard [1]. The side-jump coordinate shift is $\delta \boldsymbol{r}_{l'l} = \mathcal{A}_{l'} - \mathcal{A}_l - (\nabla_{\boldsymbol{k}} + \nabla_{\boldsymbol{k}'}) \arg V_{l'l}$ with the Berry connection $\mathcal{A} = i\langle u|\nabla u\rangle$; it is gauge invariant only with the phase of the scattering amplitude. [1] groups some τ^0 terms from asymmetric scattering with the side jump; check each paper's naming before comparing.

AVAILABLE — [2] — what the source states

Formulas for χ^{in} (the curvature dipole), χ^{sj}, $\chi^{\mathrm{sk},1}$ (intrinsic skew), $\chi^{\mathrm{sk},2}$ (extrinsic skew) of the tilted cone at leading order in the tilt; the ratios in 2.C.2 follow from them. Convention of [2]: $J^{(2)} = \chi \mathcal{E}\mathcal{E}$ for the 2ω amplitude and $J = 2\chi E^2$ at $\omega = 0$; here χ is the coefficient of E_x^2 in j_y (a factor 2), which does not affect ratios. The scaling law of [2] for one static source and its multi-source form with phonons are reproduced in 2.C.3(d).

NOT ESTABLISHED — O-theory — what 2.C does not establish

The signs of the collision terms: [3] reverses them relative to [2]. The scaling law survives both; any statement that uses the sign of a collision term (for example "the total is opposite to the dipole term") depends on the level of theory.

2.B.3. Relaxation-time, intraband, $\hbar\omega \ll 2m$. The finite-frequency behaviour of side-jump and skew terms is not given in [2] (low frequency only).

2.D.2. The average-symmetry argument follows the logic of [4]–[6] and [14]; (c) is the "statistical degeneracy" of edge states reported in [5]. The selection rule assumes that the clean part is inversion symmetric and that the steady state is a function of the Hamiltonian.

Problem 3

3.A. The zero-mode recursion and the criterion $\overline{\ln|u|} = \overline{\ln|v|}$ for the disordered chiral chain are exact [15]. It is a toy for "topology from disorder"; the experiment [5] uses long-range hoppings, van der Waals interactions and average (not exact) inversion.

3.B. B.1 is exact for any configuration. B.2 states the Niu–Thouless theorem [8]; the two-state version (charge C/q for a q-fold manifold) is the pumping analogue of the Niu–Thouless–Wu result for degenerate ground states. The displacements -0.582 and -0.418 are single-particle Wannier-centre data of the clean model at $K/J = -3/2$ (computed in the working folder).

3.C. Fig. 3.3 and the table are computed from the clean single-particle chain (24-cell ring, projected-position orbitals of bands 1–2). They are not the dressed, interacting flavors of [7]. The sign alternation of C.3 uses only $H(t+T) = H(t)$ and the flavor exchange, and holds for the interacting flavors as well.

NOT ESTABLISHED — O-calc — what 3.C does not establish

The random fields of the correlated flavor states, the effect of bond disorder (δK, δJ), and the field along the whole cycle for the interacting model.

3.D. The domain energy is an effective model consistent with [7], which states $N_c \approx \tilde{U}/(2\tilde{V}_{\max})$ and $\ell_{\min} = U/(2\tilde{V}_{\max})$; the values of E_{w}, $\beta_{\max}$, σ_h, γ, ξ are illustrative inputs, not results of [7]. The defect model of D.3 is a bookkeeping model.

NOT ESTABLISHED — O-theory — what 3.D does not establish

Any lifetime of the disordered pump. D.2 shows that the uniform gap is lost; D.3 shows which time scales with N; neither gives a nucleation rate or wall dynamics.

Appendix C. Register of available and missing results

Item	Available now	Still missing	Kind
disorder model of a real sample	projection recipe (1.A); position-disorder correlations (1.B)	defect potentials and statistics of the actual sample	O-calc
rates and broadening	Born τ_q, τ_{tr}, Γ, ξ (1.C)	vertex structure and higher cumulants for that sample	O-calc
second-order Hall coefficients	powers of τ (2.C.1); one model in two theories [2], [3]	material-specific coefficients; a consistent semiclassical theory	O-calc / O-theory
separation of mechanisms	C_1, C_2 from scaling (2.C.3)	split of C_2; independent D_x	O-calc
ensemble symmetry	selection rule $\overline{\chi} = 0$; statistical degeneracy (2.D.2)	protected transport of average SPT phases	O-theory
flavor fields of the pump	single-particle bias $0.14W$, mixing $0.01W$ (3.C.2)	dressed flavors, bond disorder, full cycle	O-calc
domains in the disordered pump	rare favourable blocks, no uniform gap (3.D.2)	nucleation rate, wall motion	O-theory
lifetime	scaling of t_{first} and t_κ with N (3.D.3)	$t_\kappa(N, W)$ for the real model	O-theory

Appendix D. Teacher notes

Provenance and checks

Every number is recomputed by `check_answers.py` (results in `check_answers.json`). Independent checks: the golden-rule rate against the exact decay of a plane wave on a 3000-site ring (0.0234 vs $0.0241\,J/\hbar$ at $W = 0.5J$); the localization length against a transfer-matrix Lyapunov exponent ($283a$ vs $288a$); $\tau_{\mathrm{tr}}/\tau_q$ against the exact angular integral (17.4 vs 18); the Berry curvature against a plaquette (Wilson-loop) computation; D_x against a two-dimensional numerical integration (-8.03 vs -8.02×10^{-4} at $v_t = 0.02$, $m = 0.4$, $\varepsilon_{\mathrm{F}} = 1$); the curvature-dipole coefficient of [2] reproduced from $-(e^3\tau/2\hbar^2)D_x$; Monte Carlo checks of cumulants, bond covariances, mean shifts, puddle averages and $p(4)$; the flavor-orbital weights and the single-period displacements from the clean five-site chain.

Interpretations to confirm

(i) The sign revision between [2] and [3] is quoted from [3]; 2.C.2(b) uses it only as a hypothetical "theory II". (ii) The body of [4] was not available; its content is represented through the original papers [5], [6]. (iii) The identification of the effective domain model of 3.D with the energetics of [7] is ours. (iv) The data of 2.C.3(b) are illustrative.

A priority list for real materials

For a low-temperature, non-magnetic crystal and a transport or second-order response, a useful first order of checks is: (1) charged impurities and trapped charges, neutral defects, vacancies and alloy disorder (on-site and bond terms, correlated); (2) displaced atoms, strain, thickness or twist-angle variations (essential for narrow bands and small gaps); (3) edges, grain boundaries and surface roughness when the sample is comparable to the mean free path; (4) phonons and other baths when the temperature or the drive supplies energy; (5) slow charge noise and control noise for long observation times (pumps); (6) magnetic impurities and domains whenever the observable depends on a magnetic symmetry. The order is set by the three tests of 1.D.3, not by the size of each term.

Sources

[1] N. Nagaosa, J. Sinova, S. Onoda, A. H. MacDonald, N. P. Ong, Rev. Mod. Phys. 82, 1539 (2010). [2] Z. Z. Du, C. M. Wang, S. Li, H.-Z. Lu, X. C. Xie, Nat. Commun. 10, 3047 (2019); arXiv:1812.08377. [3] Z. Z. Du, C. M. Wang, H.-P. Sun, H.-Z. Lu, X. C. Xie, Nat. Commun. 12, 5038 (2021); arXiv:2004.09742. [4] L. Hertkorn, Y. Oreg, Topology from disorder, Nat. Phys. 22, 1534 (2026). [5] Z. Yue et al., Average topological phase in a disordered Rydberg atom array, Nat. Phys. (2026); arXiv:2505.06286. [6] L. Su et al., Topological phase transitions and mixed-state order in a Hubbard quantum simulator, Nat. Phys. (2026); arXiv:2505.17009. [7] M. Jürgensen, J. Steiner, G. Refael, M. C. Rechtsman, Multi-band fractional Thouless pumps, arXiv:2504.09338. [8] Q. Niu, D. J. Thouless, J. Phys. A 17, 2453 (1984). [9] P. W. Anderson, Phys. Rev. 109, 1492 (1958). [10] B. Kramer, A. MacKinnon, Rep. Prog. Phys. 56, 1469 (1993). [11] S. Das Sarma, F. Stern, Phys. Rev. B 32, 8442 (1985). [12] A. A. Clerk et al., Rev. Mod. Phys. 82, 1155 (2010). [13] J. Martin et al., Nat. Phys. 4, 144 (2008). [14] R. Ma, C. Wang, Phys. Rev. X 13, 031016 (2023). [15] I. Mondragon-Shem, T. L. Hughes, J. Song, E. Prodan, Phys. Rev. Lett. 113, 046802 (2014). [16] I. Sodemann, L. Fu, Phys. Rev. Lett. 115, 216806 (2015). [17] Y. Imry, S.-k. Ma, Phys. Rev. Lett. 35, 1399 (1975).

Appendix E. Toolbox: the smallest worked examples

One entry for every tool stated under "Useful information" (or in a hint) that is above the reader's level. Each example teaches the tool, not an exam step.

Cumulants (Problem 1, Useful information)

Statement. $\mathrm{Cov}(x,y) = \overline{xy} - \overline{x}\,\overline{y}$; $\kappa_3 = \overline{(x-\overline{x})^3}$.

Smallest example. $x = 0, 1, 3$ with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$: $\overline{x} = 1$, $\mathrm{Var}\,x = \frac{1}{2}\cdot 1 + 0 + \frac{1}{4}\cdot 4 = 1.5$, $\kappa_3 = \frac{1}{2}(-1) + 0 + \frac{1}{4}\cdot 8 = 1.5$. Second way: $\overline{x^3} - 3\overline{x}\,\overline{x^2} + 2\overline{x}^3 = 7 - 7.5 + 2 = 1.5$.

Used in 1.A.2, 1.B.1, 1.B.2, 2.D.1.

Moments of a uniform variable (Problem 1, Useful information)

Statement. Uniform on $[-W/2, W/2]$: mean 0, variance $W^2/12$, $\kappa_3 = 0$.

Smallest example. $\frac{1}{W}\int_{-W/2}^{W/2} x^2 \mathrm{d}x = \frac{W^2}{12}$; the cube is odd, so $\kappa_3 = 0$. Second way: the uniform variable on $[0,1]$ has variance $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.

Used in 1.A.2, 1.C.1, 3.C.1.

Golden rule into a quasi-continuum (Problem 1, Useful information)

Statement. Rate $\frac{2\pi}{\hbar}\sum_f |V_{fi}|^2 \delta(E_f - E_i)$.

Smallest example. A level coupled with equal strength g to a ladder of levels with spacing δ: $\sum_f \delta(E_f - E_i) \to 1/\delta$, so the rate is $2\pi g^2/(\hbar\delta)$. It describes the decay for times shorter than $2\pi\hbar/\delta$; after that the ladder revives the level.

Used in 1.C.1, 1.C.3.

A Fourier sum on a ring (Problem 1, Useful information)

Statement. $\sum_{j=1}^{L} e^{i(k-k')j} = L\delta_{kk'}$.

Smallest example. $L = 3$, $k - k' = 2\pi/3$: $e^{2\pi i/3} + e^{4\pi i/3} + 1 = 0$ (the three cube roots of unity); for $k = k'$ the sum is 3.

Used in 1.C.1.

Gaussian integrals (Problem 1, Useful information)

Statement. $\int e^{-\alpha x^2/2}\mathrm{d}x = \sqrt{2\pi/\alpha}$, $\int x^2 e^{-\alpha x^2/2}\mathrm{d}x = \sqrt{2\pi/\alpha}/\alpha$.

Smallest example. $\alpha = 1$: the ratio of the two integrals is 1, the variance of a standard Gaussian. Second way: differentiate the first integral with respect to α.

Used in 1.C.2.

Noise spectrum and transition rate (Problem 1, Useful information)

Statement. $S(\omega) = \int e^{i\omega t}\overline{\eta(t)\eta(0)}\mathrm{d}t$; rate $|A_{10}|^2 S(\omega_0)/\hbar^2$.

Smallest example. For $\overline{\eta(t)\eta(0)} = \sigma^2 e^{-|t|/\tau_c}$: $S = 2\sigma^2\,\mathrm{Re}\int_0^\infty e^{(i\omega - 1/\tau_c)t}\mathrm{d}t = 2\sigma^2\,\mathrm{Re}\frac{1}{1/\tau_c - i\omega} = \frac{2\sigma^2\tau_c}{1+\omega^2\tau_c^2}$. Check: $\int S\,\mathrm{d}\omega/2\pi = \sigma^2$, the variance.

Used in 1.D.1.

Averages of a function of a random variable (Problem 1, Useful information)

Statement. $\overline{F(x)} \approx F(0) + \frac{1}{2}F''(0)\overline{x^2}$, $\mathrm{Var}\, F \approx F'(0)^2\overline{x^2}$.

Smallest example. $F = x + x^2$, Gaussian x of variance σ^2: $\overline{F} = \sigma^2$ exactly, as the formula gives; $\mathrm{Var}\, F = \sigma^2 + 2\sigma^4$, whose leading term σ^2 is the formula.

Used in 1.B.2, 1.D.2.

Berry curvature of a two-band Hamiltonian (Problem 2, Useful information)

Statement. $\Omega_\pm = \mp \boldsymbol{d} \cdot (\partial_{k_x}\boldsymbol{d} \times \partial_{k_y}\boldsymbol{d})/(2|\boldsymbol{d}|^3)$.

Smallest example. $\boldsymbol{d} = (k_x, k_y, 1)$ at $\boldsymbol{k} = 0$: $\Omega_\pm = \mp\frac{1}{2}$. Second way: $\int \Omega_- \,\mathrm{d}^2k/2\pi = \frac{1}{2}$, half of the solid angle 4π divided by 4π — the vector $\hat{\boldsymbol{d}}$ covers one hemisphere.

Used in 2.A.1.

Delta function of a function; polar integrals (Problem 2, Useful information)

Statement. $\delta(G(k)) = \delta(k - k_0)/|\mathrm{d}G/\mathrm{d}k|_{k_0}$; $\int \mathrm{d}^2k\, F(k) = 2\pi \int Fk\,\mathrm{d}k$; $\int \cos^2\phi\,\mathrm{d}\phi = \pi$.

Smallest example. $\int_0^\infty \delta(k^2 - 1)\,\mathrm{d}k = \frac{1}{2}$. $\int \mathrm{d}^2k\, k_x^2 e^{-k^2} = \pi\int_0^\infty k^3 e^{-k^2}\mathrm{d}k = \frac{\pi}{2}$; second way: $\int k_x^2 e^{-k_x^2}\mathrm{d}k_x \int e^{-k_y^2}\mathrm{d}k_y = \frac{\sqrt{\pi}}{2}\sqrt{\pi}$.

Used in 2.A.1, 2.B.2.

Integration by parts in k (Problem 2, Useful information)

Statement. $\int f\partial_{k_x} g = -\int g\partial_{k_x} f$ if $fg \to 0$.

Smallest example. One dimension, $f = e^{-k^2}$, $g = k$: $\int f\,\mathrm{d}k = \sqrt{\pi}$ and $-\int k(-2k)e^{-k^2}\mathrm{d}k = \sqrt{\pi}$.

Used in 2.B.1, 2.B.2.

Products of oscillating quantities (Problem 2, Useful information)

Statement. $\mathrm{Re}(\alpha e^{i\omega t})\mathrm{Re}(\beta e^{i\omega t}) = \frac{1}{2}\mathrm{Re}(\alpha\beta e^{2i\omega t}) + \frac{1}{2}\mathrm{Re}(\alpha\beta^*)$.

Smallest example. $\alpha = \beta = 1$: $\cos^2\omega t = \frac{1}{2} + \frac{1}{2}\cos 2\omega t$. With $\beta = i$: $-\cos\omega t \sin\omega t = -\frac{1}{2}\sin 2\omega t$ and the constant part vanishes.

Used in 2.B.3.

Heisenberg equation for fermion densities (Problem 3, Useful information)

Statement. $\dot{O} = i[H, O]$; $[c_a^\dagger c_b, n_j] = c_a^\dagger c_b(\delta_{bj} - \delta_{aj})$.

Smallest example. Two sites, $H = -J(c_2^\dagger c_1 + c_1^\dagger c_2)$: $\dot{n}_1 = -iJ(c_2^\dagger c_1 - c_1^\dagger c_2)$. For the bonding state $(|1\rangle + |2\rangle)/\sqrt{2}$ the expectation value is zero (a stationary state); for $(|1\rangle + i|2\rangle)/\sqrt{2}$ it is $-J$, a particle flowing from 1 to 2.

Used in 3.B.1.

Law of large numbers for products (Problem 3, Useful information)

Statement. $\frac{1}{n}\sum \ln|x_i| \to \overline{\ln|x|}$; $\int \ln x\,\mathrm{d}x = x\ln x - x$.

Smallest example. $x = \frac{1}{2}$ or 2 with equal probability: $\overline{x} = 1.25 > 1$, but $\overline{\ln x} = 0$, so a long product neither grows nor decays exponentially — its typical size is set by the logarithmic mean. And $\int_0^1 \ln x\,\mathrm{d}x = -1$.

Used in 3.A.1, 3.A.2.

Pauli decomposition (Problem 3, Useful information)

Statement. $M = h_0\mathbb{1} + \boldsymbol{h}\cdot\boldsymbol{s}$ with $h_z = (M_{11} - M_{22})/2$, $h_x - ih_y = M_{12}$.

Smallest example. $M = \begin{pmatrix} 3 & 1-i \\ 1+i & 1 \end{pmatrix}$: $h_0 = 2$, $h_z = 1$, $h_x = h_y = 1$. Second way: the eigenvalues $h_0 \pm |\boldsymbol{h}| = 2 \pm \sqrt{3}$ equal those of M.

Used in 3.C.1, 3.C.2.

Gaussian sums and Φ (Problem 3, Useful information)

Statement. A sum of ℓ independent Gaussians of variance σ^2 has variance $\ell\sigma^2$; $\Phi(-1) = 0.159$.

Smallest example. Four standard Gaussians: the sum has rms 2, so $P(\text{sum} > 2) = \Phi(-1) = 0.159$.

Used in 3.D.2.

Independent rare events (Problem 3, Useful information)

Statement. Probability of no event in time t: $e^{-\Lambda t}$.

Smallest example. Two independent sources, each at rate $\Lambda/2$: no event in either, $e^{-\Lambda t/2}e^{-\Lambda t/2} = e^{-\Lambda t}$ — rates of independent sources add.

Used in 3.D.3.

End of solution booklet